

BioScreen

Function-based DNA synthesis screening and actuarial pricing for biosecurity

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The problem

- DNA synthesis screening checks orders by sequence similarity to known toxins.
- AI protein design can now rewrite a toxin's sequence while keeping its function.
- Microsoft-led 2025 red-teaming study: up to 100% of AI-designed ricin variants evaded major screening tools.
- The same weakness enables order fragmentation and benchtop synthesizers.

Why it matters at 100 Days Mission speed

- Rapid vaccine development runs on rapid DNA synthesis.
- The same AI tools that accelerate legitimate design also accelerate misuse.
- Current screening systems were built for a slower, less automated world.
- The gap is widest in LMIC settings where screening infrastructure is thinnest.

The solution - Layer 1: functional screening

- ESM-C 600M protein language model embeddings (open weights, MIT license).
- Contrastive per-residue alignment: hazard = similarity to toxins minus similarity to safe proteins.
- Catches rewritten toxins by what they do, not what they look like.
- Runs on a laptop. No proprietary data. Free to use.

The solution - Layer 2: actuarial pricing

- Screening output becomes a risk score, priced as a premium on the order.
- A 0.12% loading on synthesis orders funds a global biosecurity pool.
- Verified customers pay 65% less - transparency becomes profitable.
- Pool funds LMIC screening subsidies; mechanism is self-financing.

What we've validated

- Sensitivity: 100% - 16/16 controlled toxins flagged.
- Specificity: 100% - 0/4 safe proteins flagged.
- Toxins: ricin, botulinum A/B/E, shiga 1/2, diphtheria, abrin, modeccin, anthrax complex, tetanus.
- Safe references: GFP, lysozyme, insulin, BSA.
- Separation margin: 0.0393. Documented limitation: needs a structural layer for production margin.
- Pricing: 5,000-order simulation extrapolates to ~\$15M/year globally.

The economic mechanism

- Today: screening costs fall on providers. No incentive to invest.
- With BioScreen: risk is priced. High-risk orders pay more. Verified customers pay less.
- Incentive-compatibility: 65% premium savings for verified customers.
- Funding scales with adoption - no external subsidy required.

Who uses it and who pays

Users: DNA synthesis providers and the labs that order from them.

Payers: The same group. A 0.12% loading on every order.

- Large providers absorb it as compliance overhead.
- Smaller labs pass it through. Verified labs pay 65% less.
- The \$50K award builds and validates the prototype. It does not fund operations.

Dual-use and misuse

- Hazard: the method itself could be used as an oracle to iterate around.
- Mitigations: complementary to sequence screening, not a replacement.
- Per-provider thresholds. Managed signature set (like antivirus).
- Human review for high-risk orders; no automated accept/reject.
- We publish adversarial test results ourselves - honest limits documented.

What we build in six months

- Widen separation margin from 0.039 to above 0.15 via a structural similarity layer (ESMFold + FoldSeek).
- Expand reference database from 16 to 100+ controlled toxins.
- Publish a benchmark of AI-designed toxin variants - reproducible, open.
- Complete one industry pilot with a synthesis provider or CRO.
- Submit an open preprint to bioRxiv.
- Calibrate pricing against real order data from at least one partner.

How the \$50,000 is spent

Personnel - \$30,000. One full-time research engineer, six months, Kenya salary scale.

GPU compute - \$5,000. ESMFold predictions and embedding at scale.

Wet-lab validation - \$7,500. Synthesis of test panels for in vitro ground-truth.

Partner engagement - \$4,000. Site visits, data-sharing agreements, pilot setup.

Publication & dissemination - \$2,000. Open-access fees, one conference.

Contingency - \$1,500.

Team and ask

Peter Mburu Kariuki - Founder

B.Tech in Biotechnology, Technical University of Kenya. M.Sc. in Financial Engineering (in progress), WorldQuant University. IT Project Manager, Equity Group Foundation.

What we need from BioPREVAIL beyond funding

- Industry introductions - one synthesis provider for a pilot.
- Regulatory guidance - WHO, BWC ISU, or national biosafety authorities.
- Scientific critique - reviewers who will stress-test the method.

Code and validation results: github.com/Petermburu-star/BioScreen

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